EXISTENCE OF INJECTIVE RESOLUTIONS UNDER ENOUGH INJECTIVES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 6511-6522, following the statement at 6496-6509.

Editorial note. The recursion is displayed for the first two steps and repeated verbatim thereafter ('And so on'), matching the established induction-repetition note practice.

Allowed standard prerequisites (editorial).

- Enough injectives. hypothesis; used at each step of the recursion. Embeds quotients into injective objects.
- Right resolutions against subcategories. lemma-subcategory-right-resolution (same chapter, cited at ~6522). Part (3) is its special case.

Source reference. Original lemma 013J: sources/stacks-derived/pinned-original-source.tex, source lines 6476–6476. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05T6: sources/stacks-derived/pinned-original-source.tex, source lines 5604–5604. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Assume $\mathcal{A}$ has enough injectives.*

(1) *Any object of $\mathcal{A}$ has an injective resolution.*
(2) *If $H^n(K^\bullet) = 0$ for all $n \ll 0$ then $K^\bullet$ has an injective resolution.*
(3) *If $K^\bullet$ is a complex with $K^n = 0$ for $n < a$, then there exists an injective resolution $\alpha : K^\bullet \to I^\bullet$ with $I^n = 0$ for $n < a$ such that each $\alpha^n : K^n \to I^n$ is injective.*

Proof. Proof of (1). First choose an injection $A \to I^0$ of A into an injective object of $\mathcal{A}$. Next, choose an injection $I_0/A \to I^1$ into an injective object of $\mathcal{A}$. Denote d^0 the induced map $I^0 \to I^1$. Next, choose an injection $I^1/\operatorname{Im}(d^0) \to I^2$ into an injective object of $\mathcal{A}$. Denote d^1 the induced map $I^1 \to I^2$. And so on. By Lemma 013J part (2) follows from part (3). Part (3) is a special case of Lemma 05T6. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.